Strong normalization proof strategies for typed λ-calculi

New decreasing measures

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CONICET

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Facultad de Ciencias Exactas y Naturales - UBA

ICC Instituto de Ciencias de la Computación

Proving termination in programming languages

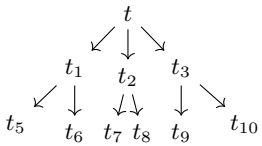
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Simply typed λ-calculus is terminating

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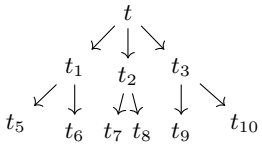
Reduction tree



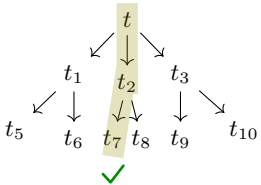
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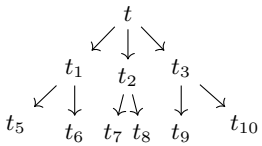
Weak normalization WN



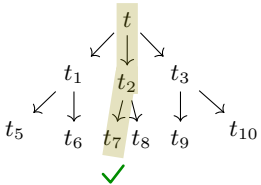
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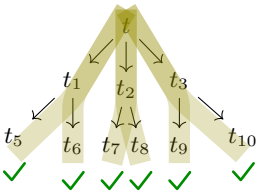
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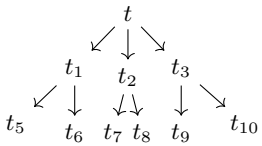
Strong normalization (SN)



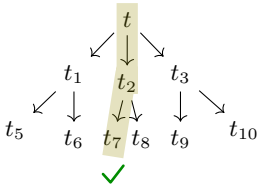
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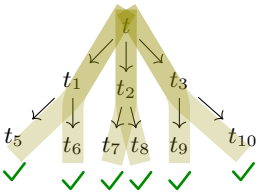
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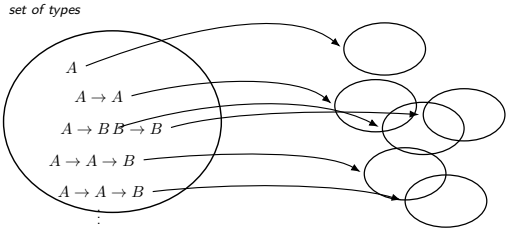
Strong normalization (SN)



Proving SN usual

- safest system/strongest result
- logical consistency (via Curry-Howard)
- hard (and beautiful!) problem

Reducibility

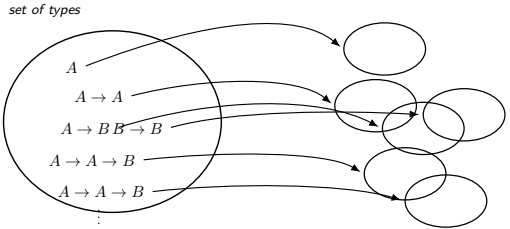


$$[\![A \to B]\!] = \{t \in A \to B \mid \forall s \in [\![A]\!].ts \in [\![B]\!]\}$$

$$\mathit{CR1}([\![A]\!]) \qquad \mathit{CR2}([\![A]\!]) \qquad \mathit{CR3}([\![A]\!])$$

$$\Gamma \vdash t : A \implies t \in [\![A]\!]$$

Reducibility



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$CR1([\![A]\!])$ $CR2([\![A]\!])$ $CR3([\![A]\!])$

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abstraction

concise **lacking intuition**

adaptable **why it works?**

Gallier'94

This paper provides some answers to the above questions. But before explaining our results, we would like to explain our motivations and our point of view a little more. Reducibility proofs are seductive and thrilling, but also elusive. Following these proofs step-by-step, we see that they "work" (when they are not wrong!), but I claim that most of us would still admit that they are not sure *why* these proofs work! The situation is somewhat comparable to driving a Ferrari (I suppose): the feeling of power is tremendous, but what exactly is under the hood? What kind of carburator, what kind of valve mechanism, gives such power and flexibility?

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de Vrijer'87

The *loose valuation* only gives an upper bound for the height. But the proof of strong normalization that corresponds to it—here called the "quick proof"—is simple and transparent. Presumably more transparent than the ones involving a computability predicate. As a matter of fact the quick proof can be viewed as providing some extra insight into the nature of the computability proofs. The computability predicate should then be analyzed as abstracting from all the information—still explicit in the labeled functionals—that is not strictly needed in an induction on terms which successfully establishes strong normalization.

Decreasing measures

Decreasing measure

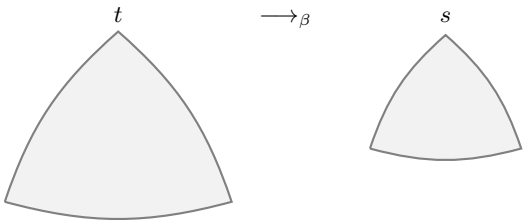
$\#(\cdot)$: λ-terms $\to$ well-founded order $\qquad t \to_\beta s \implies \#(t) > \#(s)$

Decreasing measures

Decreasing measure

$\#(\cdot) : \lambda\text{-terms} \to \text{well-founded order} \qquad t \to_\beta s \implies \#(t) > \#(s)$

Intuitively

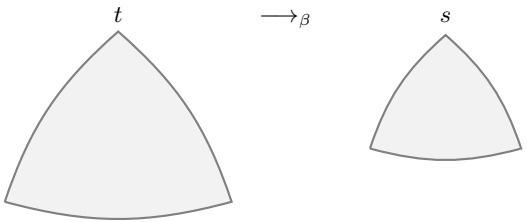


Decreasing measures

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Intuitively



Why?

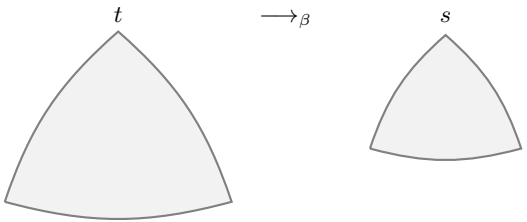
- Information
- Insight

Decreasing measures

Decreasing measure

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Intuitively



Why?

- Information
- Insight

Existing measures

- Gandy'80
- de Vrijer'87

translation to λ-I

+

interpretation on HSMF

(hereditarily strictly monotonic functions)

Contributions of this thesis

Four decreasing measures for typed λ-calculi

1. 3 for STLC
2. 1 (and a half) for IIT

Gödel's koan (open question) [TLCA #26, RTA #19]

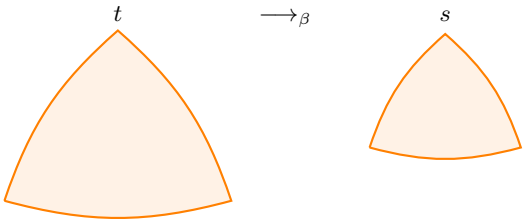
Gödel's koan (open question) [TLCA #26, RTA #19]

To find an "easy" mapping from simply typed λ–calculus to ordinals

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To find an "easy" mapping from simply typed λ–calculus to ordinals



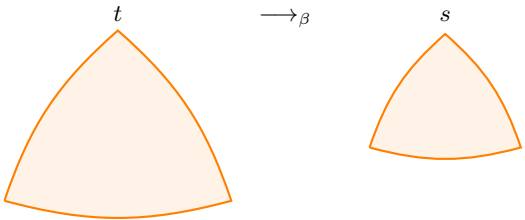
Koan?

- "Easy" is not well defined
- Solutions space unclear
- A solution should be clearly recognizable

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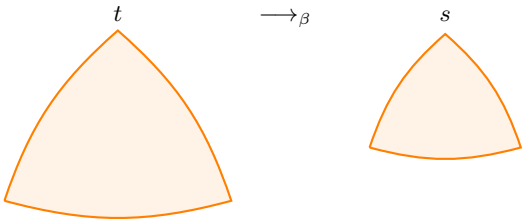
Complexity dimensions

- definition
- proof
- understand
- compute

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[TLCA #26, RTA #19]

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Complexity dimensions

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- proof
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The koan in this thesis

- Long-term motivation (not a quickly solvable question)
- Easy at some dimensions, but not all
- We contributed towards possible solutions
- We did not solve it

$\mathcal{W}$: A simple decreasing measure for STLC

[Barenbaum & Sottile, FSCD'23]

Motivation

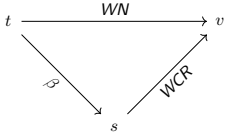
- wanted a somehow easier DM
- concretized one from known ideas

DECrease from INCrease

Nederpelt-Klop's lemma $\quad INC \wedge WCR \wedge WN \implies SN \wedge CR$

DECrease from INCrease

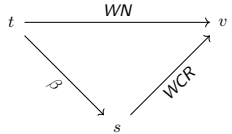
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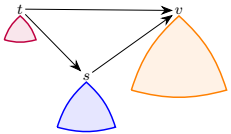
$$INC \wedge WCR \wedge WN \implies SN \wedge CR$$

DECrease from INCrease

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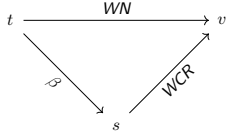


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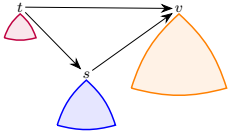


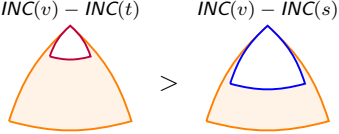
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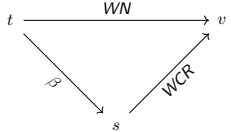
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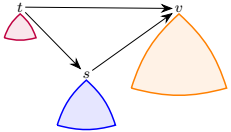


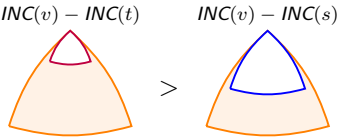
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The memory calculus $\lambda^{\mathbf{m}}$

Avoid erasure

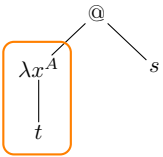
$\cancel{(\lambda x.y)z \to_\beta y}$

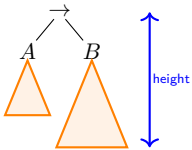
$(\lambda x.y)z \to_{\mathbf{m}} y\langle z\rangle$

- Known idea (Church, Nederpelt, Klop)
- INC (more memories)
- WCR (by usual means)

Redex degrees and Turing's observation

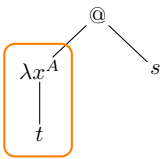
Degree of a redex

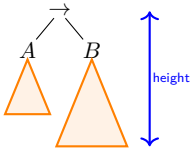




Redex degrees and Turing's observation

Degree of a redex





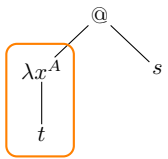
Redex creation example [Lévy]

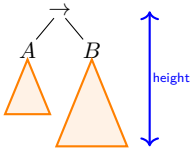
$(\lambda x.\ \ldots\ xu\ \ldots)(\lambda y^A.s) \longrightarrow_\beta \ \ldots\ (\lambda y^A.s)u\ \ldots$

$\mathsf{h}((A \to B) \to C)$ $\qquad$ $\mathsf{h}(A \to B)$

Redex degrees and Turing's observation

Degree of a redex





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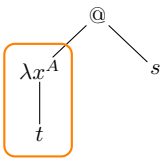
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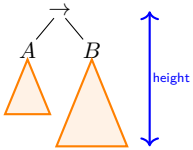
Turing's observation

The degree of a ***created*** *redex*
is smaller than its ***creator***
(*Any redex can copy any degree*)

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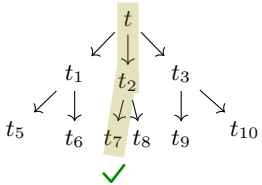
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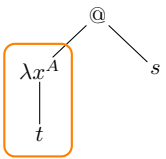
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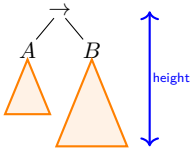
A *weak* DM [Turing, Gandy, Prawitz]



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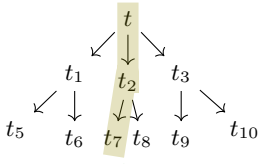
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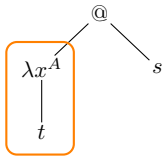


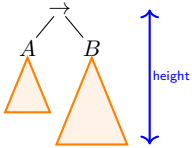
Multiset of degrees

$[d, d, d, \ldots] \to_\beta [d, d, \ldots]$

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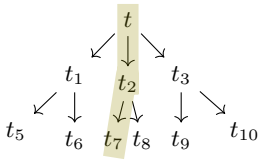
$\mathsf{h}((A \to B) \to C)$ $\qquad$ $\mathsf{h}(A \to B)$

Turing's observation

The degree of a ***created*** *redex is smaller than its* ***creator***

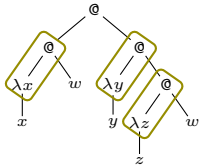
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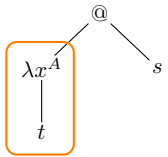
Multiset of degrees

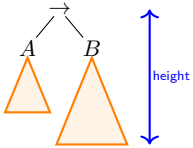
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Redex degrees and Turing's observation

Degree of a redex





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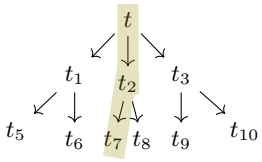
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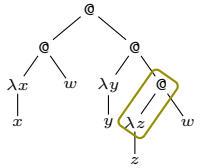
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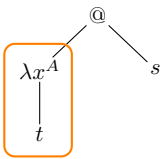
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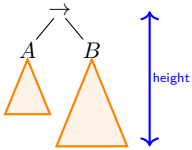
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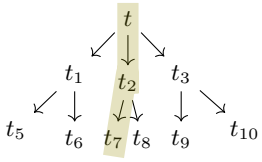
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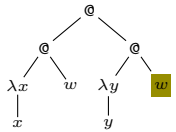
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✓

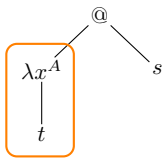
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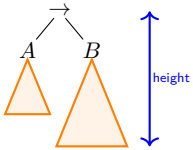
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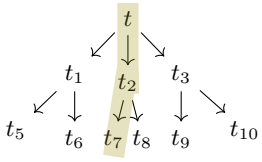
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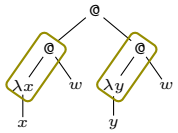
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Simplification: obtaining the normal form in λ^{m}

Simplification by degree

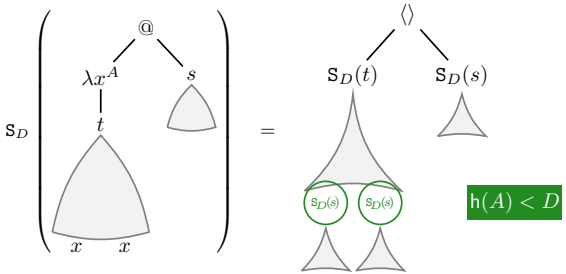
Simulate the execution of all d-redexes

Simplification: obtaining the normal form in λ^{m}

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$\mathtt{s}_D((\lambda x^A.t)s) :=$ (with deg $= D$)



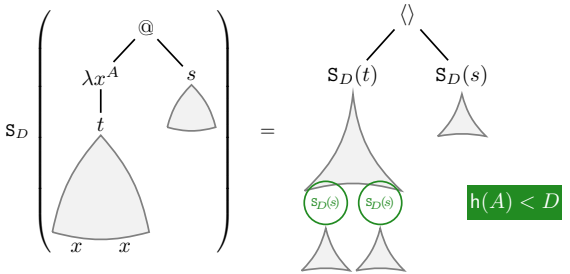
$\mathsf{h}(A) < D$

Simplification: obtaining the normal form in $\lambda^{\mathbf{m}}$

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1. $\mathtt{S}_d$ sound

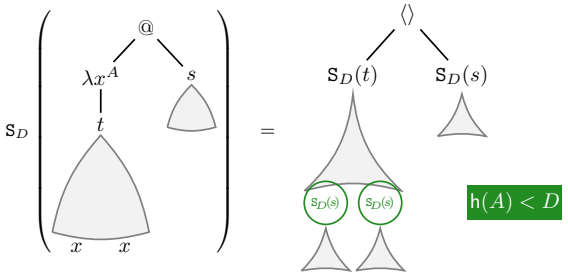
$t \to^*_{\mathbf{m}} \mathtt{S}_d(t) \in \lambda^{\mathbf{m}}$

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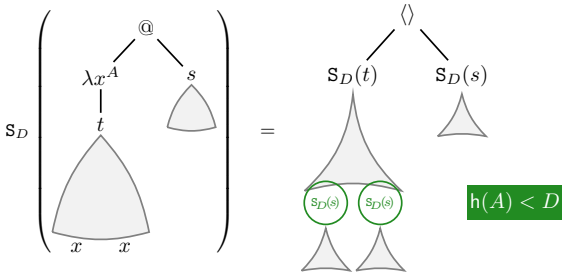
1. $\mathtt{S}_d$ sound — $t \to^*_{\mathbf{m}} \mathtt{S}_d(t) \in \lambda^{\mathbf{m}}$
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Full simplification

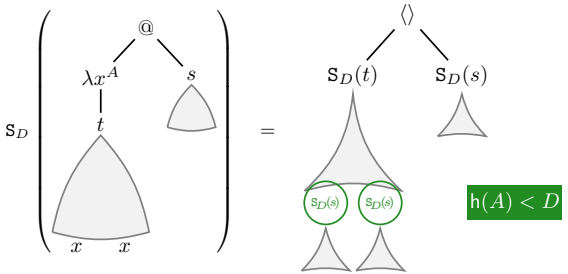
Apply d-simplification from D to 1

Simplification: obtaining the normal form in λ^{m}

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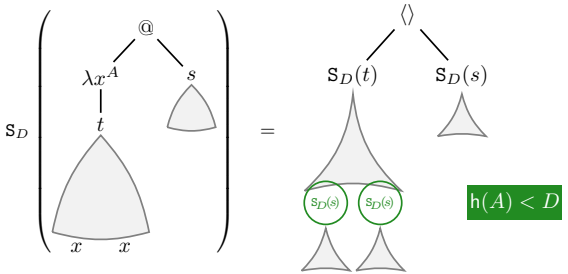
1. $\mathtt{S}_*$ normal form — $\mathtt{S}_*(t) = \Downarrow t$

Simplification: obtaining the normal form in $\lambda^{\mathtt{m}}$

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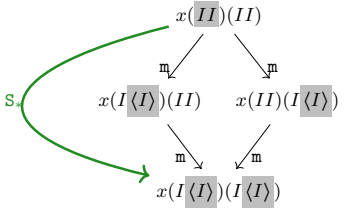
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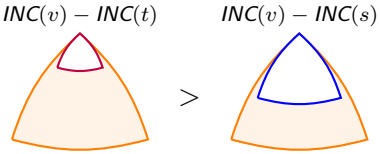
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Intuition

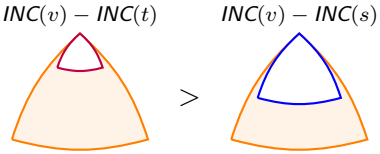


Measure $\mathcal{W}$: definition



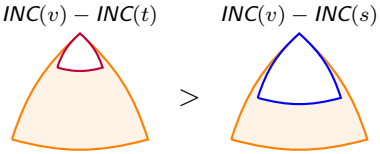
$$\mathsf{w}(\mathsf{S}_*(t)) - \mathsf{w}(t)$$

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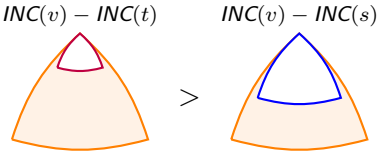
Not a STLC measure... but a $\lambda^{\mathfrak{m}}$ one

Measure $\mathcal{W}$: definition



$$\mathsf{w}(\mathsf{S}_*(t)) - \mathsf{w}(t)$$

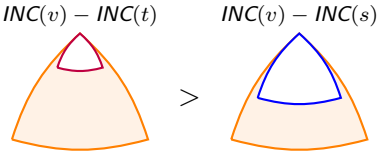
Measure $\mathcal{W}$: definition



$\lambda^{\rightarrow} \subseteq \lambda^{\mathbf{m}}$ $\quad \mathsf{w}(M) = 0$

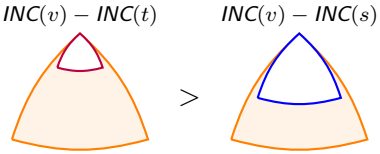
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$\mathsf{w}(\mathbf{S}_*(M)) - 0$

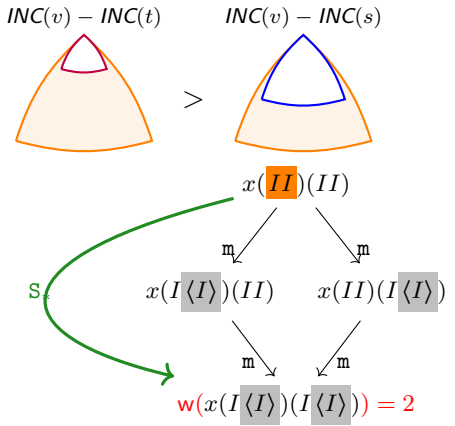
Measure $\mathcal{W}$: definition



$\mathcal{W}$: a STLC measure

$\mathcal{W}(M) = \mathsf{w}(\mathbf{S}_*(M))$

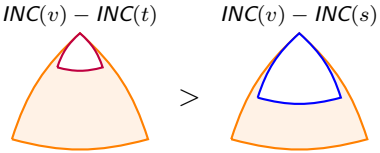
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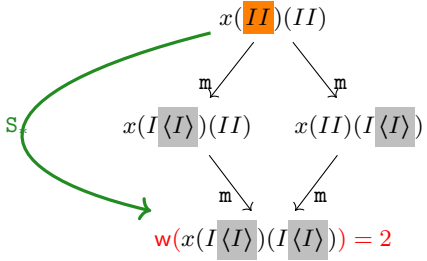
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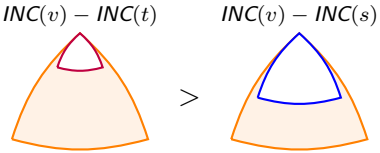
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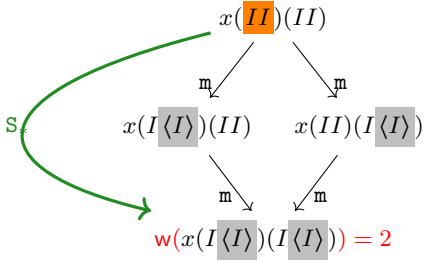
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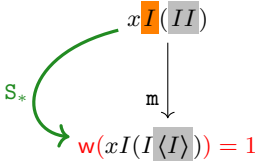


$\mathcal{W}$: a STLC measure

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Measure $\mathcal{W}$: decrease proof

Main theorem

$$M \to_\beta N \qquad \Longrightarrow \qquad \mathcal{W}(M) > \mathcal{W}(N)$$

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Up to now...

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Forgetful reduction

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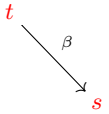
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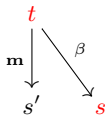
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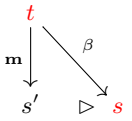
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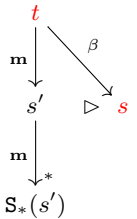
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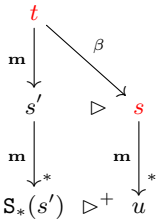
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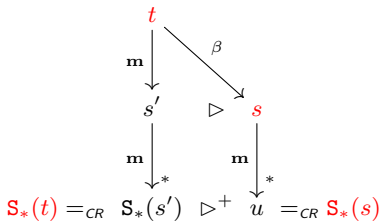
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Quick conclusion

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Turing's observation

Nederpelt-Klop's lemma ⇓

Klop's non-erasure ⇓

Gandy's strong DMs

⇓

purely syntactic strong DM

Quick conclusion

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$\Downarrow$ $\Downarrow$ $\Downarrow$

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Strong measures [Gandy, de Vrijer]

Translation to λ-I + interpretation to HSMF

proof based on HSMF properties vs λ–calculus properties

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$\mathcal{W}_\cap$: A simple measure for IIT

[Barenbaum, Ronchi Della Rocca & Sottile, MFPS'25]

Motivation

- Koan
- Provide a simple proof
- No native strong DM

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Central idea: **adapt $\mathcal{W}$ to IIT**

define simplification

$$\mathtt{S}_D((\lambda x^A.t^B)s) \;=\; \mathtt{S}_D(t)[\mathtt{S}_D(s)/x]\langle \mathtt{S}_D(s)\rangle \qquad \textbf{if } \deg(A \to B) = D$$

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S_D deals with **derivations** $\;\therefore\;$ we need Church-style terms

Church-style with parallel reductions

[Liquori & Ronchi Della Rocca]

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$M ::= ... \mid M\,M$

$t ::= ... \mid t\{t_1, ..., t_n\}$

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$\mapsto$

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Parallel reductions

$$(\lambda x.t)\{t_1, ..., t_n\} \to_\beta (\lambda x.t)\{t'_1, ..., t'_n\}$$

Invariant: all terms in a $\{t_1, ..., t_n\}$ are equal modulo type erasure.

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Problem incompatible with simplification by degree ($\mathtt{S}_d$)

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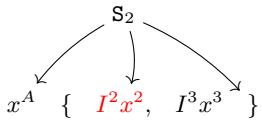
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different types $\implies$ different redex degrees

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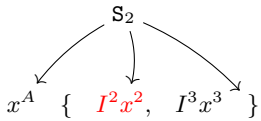
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we need a system accounting for this "intermediate" reduction information

$\Lambda^{\mathsf{Ch}}_\cap$: Idempotent intersection types à la Church

Linear reduction Inspired by Kfoury's

$$\begin{array}{ccc} M & \xrightarrow[\mathsf{Cu}]{} & N \\ \sqcup & & \sqcup \end{array}$$

$$\{t_1, ..., t_n\} \xrightarrow[\mathsf{Ch}]{} \{s_1, ..., t_n\} \xrightarrow[\mathsf{Ch}]{} ... \xrightarrow[\mathsf{Ch}]{} \{s_1, ..., s_n\}$$

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Injectivity in grammar and typing rules

$\{A_1, ..., A_n\}(A_i \neq A_j \text{ if } i \neq j)$ $\quad$ $\{t_1, ..., t_n\}(t_i \neq t_j \text{ if } i \neq j)$ $\quad$ $\dfrac{(\Gamma \vdash_{\mathsf{Ch}} s_i : A_i)_{i \in 1..n} \qquad A_i \neq A_j \text{ if } i \neq j}{\Gamma \Vdash_{\mathsf{Ch}} \{s_1, \ldots, s_n\} : \{A_1, \ldots, A_n\}}$

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Substitution parallel

$$(\lambda x^{\{A_1,...,A_n\}}.t)\{s_1^{A_1},...,s_n^{A_n}\}$$

$$\downarrow_{\mathsf{Ch}}$$

$$t[s_1^{A_1}/x^{A_1},\dots,s_n^{A_n}/x^{A_n}]$$

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Bijection between $\vec{s}$ and $\vec{A}$ in $\Gamma \Vdash \vec{s} : \vec{A}$ $\quad$ $\Rightarrow$) by type uniqueness $\quad$ $\Leftarrow$) by injectivity

$\Lambda_\cap^{\mathsf{Ch}}$ with memories: $\Lambda_\cap^{\mathsf{Chm}}$

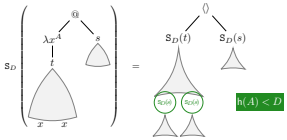
Replicated λ^{m} development in $\Lambda_\cap^{\mathsf{Ch}}$

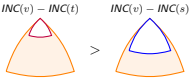
$$(\lambda x^{\{A_1,\ldots,A_n\}}.y)\{x_1,\ldots,x_n\} \to_{\mathbf{m}} y\langle\{x_1,\ldots,x_n\}\rangle$$

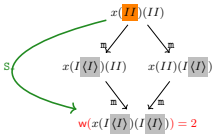
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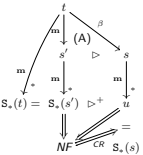
Replicated $\lambda^{\mathfrak{m}}$ development in $\Lambda_\cap^{\mathsf{Ch}}$

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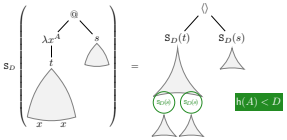


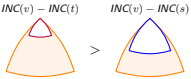


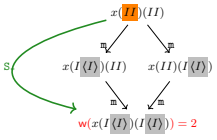
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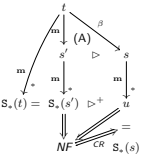
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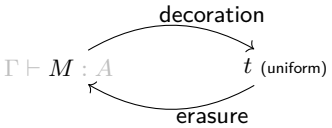


But...

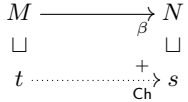
- $\mathcal{W}_\cap$ proves SN for $\Lambda_\cap^{\mathsf{Ch}}$
- we want a proof for $\Lambda_\cap^{\mathsf{Cu}}$

Lifting the measure to $\Lambda_\cap^{\mathsf{Cu}}$

Correspondence



Simulation



Curry-style measure the measure of the Church-correspondent

Main theorem

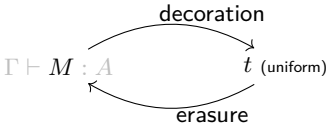
$$M \to_\beta N \implies \mathcal{W}_\cap(M) > \mathcal{W}_\cap(N)$$

Completing the picture: typable iff SN

$$\Gamma \vdash M : A \iff M \in SN$$

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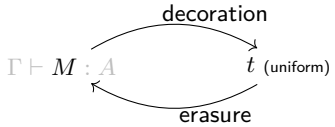
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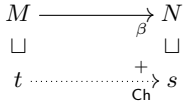
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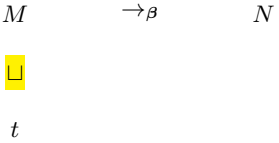
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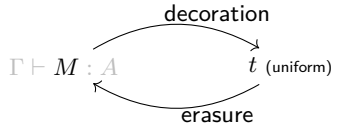
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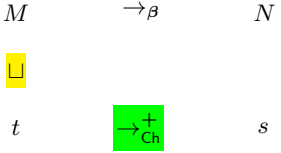
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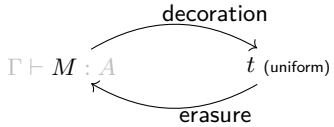
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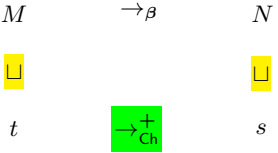
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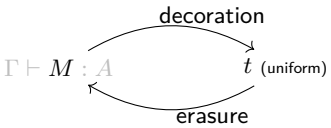
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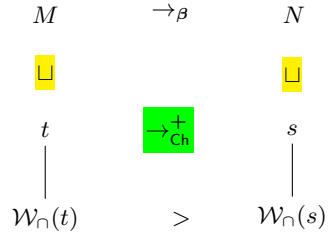
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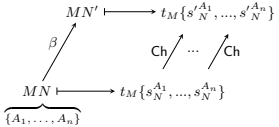
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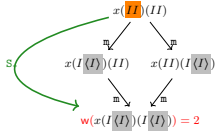
$$\Gamma \vdash M : A \iff M \in SN$$

Quick conclusion and related works

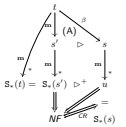
Church-style IIT



First native strong DM

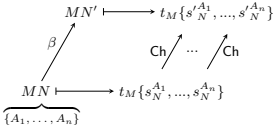


Simplest codomain: $\mathbb{N}$

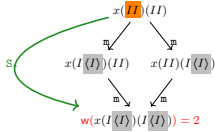


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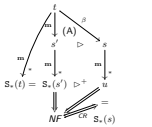
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Native measures for IIT [Kfoury & Wells'95, Boudol'03]

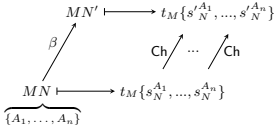
multiset vs pair vs $\mathbb{N}$

weak (strategy) vs strong (arbitrary)

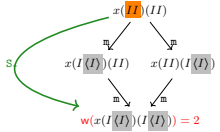
Church-style proven correspondent

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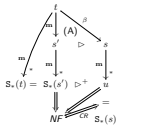
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Church-style proven correspondent

Translation based: from IIT to STLC [Bucciarelli, Piperno & Salvo]

$$[\![\lambda x^{A,B}.t]\!] \mapsto \lambda x_1^A.\lambda x_2^B.[\![t]\!]$$

Straigthforward
composing proved results

Could be defined
the paper is about the translation

Domain shift
loses intersection type details

Finer analysis is not possible $t \in \Lambda_\cap \backslash STLC \implies$ more reduction steps after translation

$\hat{\mathcal{W}}$: exact refinement for $\mathcal{W}$

Motivation

1. $\mathbb{N}$ measures $\implies$ upper bound for reductions length
2. Tightness of the upper bound

Is the upper bound exact?

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Central idea: **refine $\mathcal{W}$ to exactness**

Fixing overmeasurement

Exact memory calculus $\lambda^{\hat{\mathfrak{m}}}$

Problem we always remember

$(\lambda x.y)s \longrightarrow_\beta y$

Fixing overmeasurement

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$$(\lambda x.y)s \longrightarrow_\beta y \qquad (\lambda x.y)s \longrightarrow_\beta y\langle s\rangle$$

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Solution only remember **on erasure**

$$(\lambda x.y)s \longrightarrow_\beta y \qquad (\lambda x.y)s \longrightarrow_\beta y\langle s\rangle \qquad (\lambda x.x)s \longrightarrow_\beta s\langle s\rangle$$

$$(\lambda x.t)s \xrightarrow{x\in \mathsf{fv}(t)} t[s/x]\langle\rangle \qquad (\lambda x.t)s \xrightarrow{x\notin \mathsf{fv}(t)} t\langle s\rangle$$

Fixing overmeasurement

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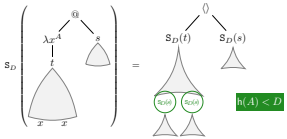
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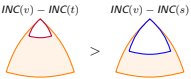
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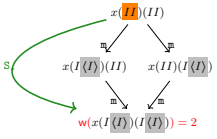
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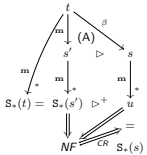
The measure $\hat{\mathcal{W}}$

Simplification and properties









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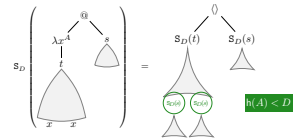
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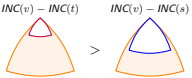
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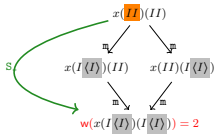
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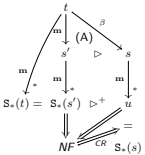
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Definition

$$\hat{\mathcal{W}}(t) \;=\; \mathsf{w}(\mathsf{S}_*(t))$$

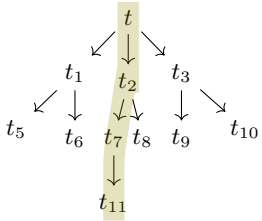
Decrease proof

$$M \to_\beta N \qquad \Longrightarrow \qquad \hat{\mathcal{W}}(M) > \hat{\mathcal{W}}(N)$$

Exactness: the proof

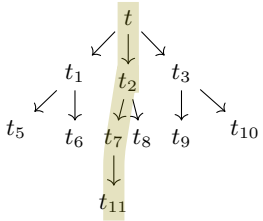
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Maximal strategy F_∞ [van Raamsdonk]



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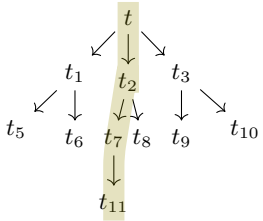
Enables LRC length $|M|$

$$(\lambda x.M)N \xrightarrow{x \in \mathrm{fv}(t)} |M[N/x]\vec{P}| + 1$$

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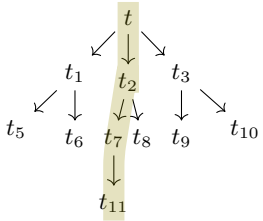
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Decompose $\mathcal{W}$ calculation

$$\begin{aligned}
\hat{\mathcal{W}}(t\langle s\rangle) &= \hat{\mathcal{W}}(t) + \hat{\mathcal{W}}(s) + 1 \\
\hat{\mathcal{W}}(t\langle s\rangle u) &= \hat{\mathcal{W}}((tu)\langle s\rangle) \\
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$$

Exactness theorem

$$\hat{\mathcal{W}}(M) = |M|$$

- By induction on ISN [van Raamsdonk]

$$\frac{N_1 \in \mathsf{ISN} \quad \dots \quad N_n \in \mathsf{ISN}}{x N_1 \dots N_n \in \mathsf{ISN}}\ \mathsf{ISN}_x \qquad \frac{M \in \mathsf{ISN}}{\lambda x.M \in \mathsf{ISN}}\ \mathsf{ISN}_\lambda$$

$$\frac{M[N/x]\, P_1 \dots P_n \in \mathsf{ISN} \quad N \in \mathsf{ISN}}{(\lambda x.M)\ N\ P_1 \dots P_n \in \mathsf{ISN}}\ \mathsf{ISN}_@$$

- Relate $\hat{\mathcal{W}}(M)$ and $|M|$ through the decomposition rearrangement

Quick conclusion and ongoing work

Quantitative approach for $\mathcal{W}$

- What information does the number provide?
- Can we get more information from it?

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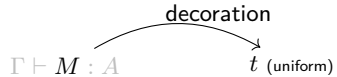
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weakening in



$\Longrightarrow$ $|M| > |t|$

Measure $\mathcal{T}^{\mathbb{m}}$: motivation

Motivation extend Turing's result

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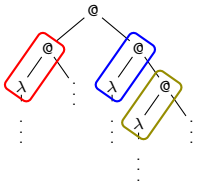
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Original measure

Term t



Multiset

$$\mathcal{T}(t) \quad = \quad [\ 3\ ,\ 2\ ,\ 1\ ]$$

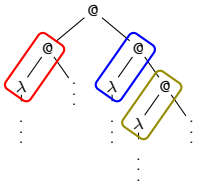
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Counterexample for arbitrary redex decrease

$$\underbrace{(\lambda x^\tau ....x...x...)(\underbrace{Ky^\tau}_{2}y^\tau)}_{1} \quad \to_\beta \quad ...(\underbrace{Ky^\tau}_{2}y^\tau)...(\underbrace{Ky^\tau}_{2}y^\tau)...$$

$$[2,1] \quad < \quad [2,2]$$

Collecting more info

Change ideas

1. Family indexed by degrees
   only redexes up to d
   $\mathcal{T}^{\mathbb{m}}_3(t),\ \mathcal{T}^{\mathbb{m}}_2(t),\ \mathcal{T}^{\mathbb{m}}_1(t)$

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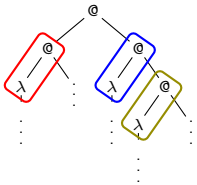
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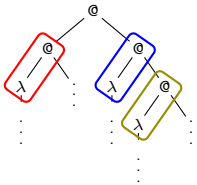
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Reduction (restricted) by degrees

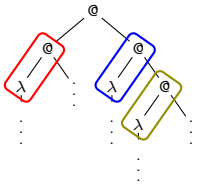
$$t \xrightarrow{d} s$$

Properties

- Confluence
- Strong normalization

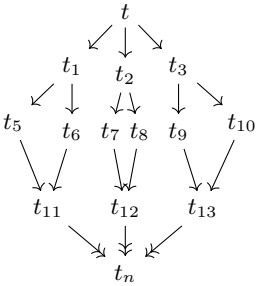
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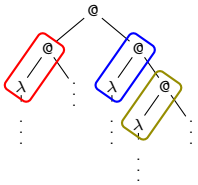
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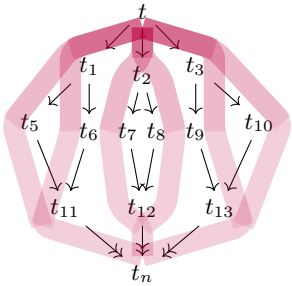
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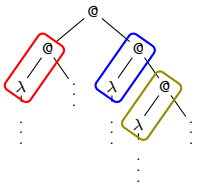
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   e.g., measure for lower degree $(d, \mathcal{T}^{\mathfrak{m}}_{d-1})$

Intuition

Term t



Multiset

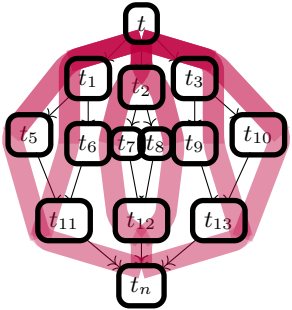
$$\mathcal{T}_3(t) = [\ (3,?)\ ,\ (2,?)\ ,\ (1,?)\ ]$$
$$\mathcal{T}_2(t) = [\ (2,?)\ ,\ (1,?)\ ]$$
$$\mathcal{T}_1(t) = [\ (1,?)\ ]$$

Reduction (restricted) by degrees

$$t \xrightarrow{d} s$$

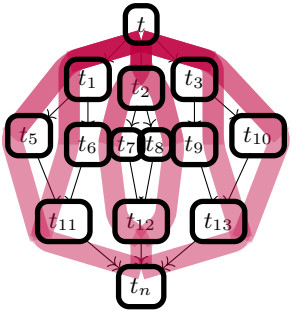
Properties

- Confluence
- Strong normalization



Measure $\mathcal{T}^{\mathbb{m}}$: collecting THE info

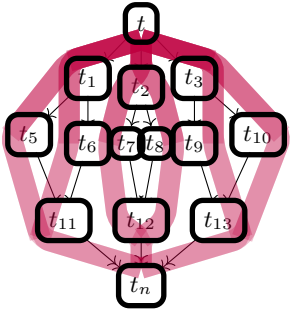
1. Family indexed by degrees
   only redexes up to d



2. Collect more information
   (redex degree, ???)

Measure $\mathcal{T}^{\mathbb{m}}$: collecting THE info

1. Family indexed by degrees
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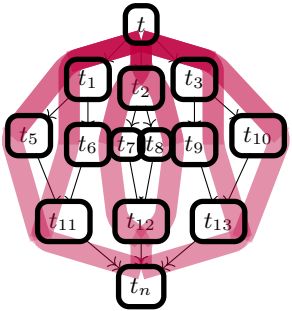


$[\quad \mathcal{T}_{d-1}$

2. Collect more information
   (redex degree, ???)

Measure $\mathcal{T}^{\mathbb{m}}$: collecting THE info

1. Family indexed by degrees
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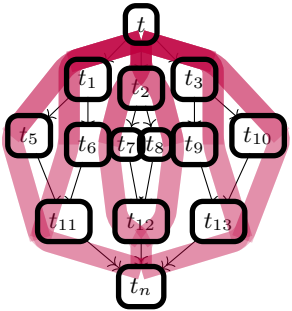


$[\quad \mathcal{T}_{d-1}(t)$

2. Collect more information
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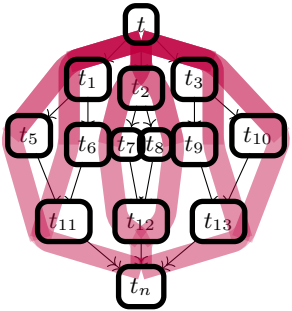


2. Collect more information
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$$[\quad \mathcal{T}_{d-1}(t) \quad , \quad \mathcal{T}_{d-1}(t_1)$$

Measure $\mathcal{T}^{\mathbb{m}}$: collecting THE info

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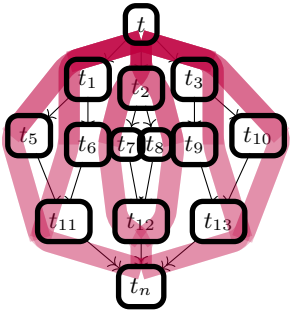


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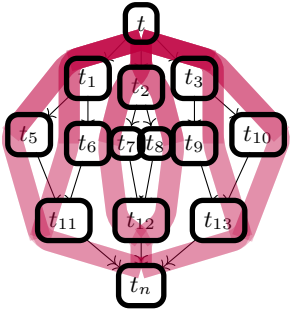


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$$[\quad \mathcal{T}_{d-1}(t) \quad , \quad \mathcal{T}_{d-1}(t_1) \quad , \quad \dots \quad , \quad \mathcal{T}_{d-1}(t_n) \quad]$$

Measure $\mathcal{T}^{\mathbb{m}}$: collecting THE info

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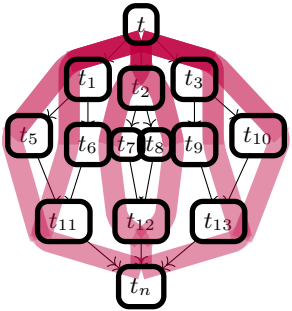


2. Collect more information
   (redex degree, ???)

$$\underbrace{[\ \mathcal{T}_{d-1}(t)\ ,\ \mathcal{T}_{d-1}(t_1)\ ,\ \dots\ ,\ \mathcal{T}_{d-1}(t_n)\ ]}_{[\ \mathcal{T}^{\mathbb{m}}_{d-1}(t')\ ||\ \rho : t \xrightarrow{d}{}^{*}_{\mathbf{m}} t'\ ]}$$

Measure $\mathcal{T}^{\mathfrak{m}}$: collecting THE info

1. Family indexed by degrees
   only redexes up to d



2. Collect more information
   (redex degree, ???)

$$\underbrace{[\ \mathcal{T}_{d-1}(t)\ ,\ \mathcal{T}_{d-1}(t_1)\ ,\ \dots\ ,\ \mathcal{T}_{d-1}(t_n)\ ]}_{[\ \mathcal{T}^{\mathfrak{m}}_{d-1}(t')\ ||\ \rho : t \xrightarrow{d}{}^{*}_{\mathbf{m}} t'\ ]}$$

Definition

$$\mathcal{T}^{\mathfrak{m}}_D(t) = [\ (d, [\ \mathcal{T}^{\mathfrak{m}}_{d-1}(t')\ ||\ \rho : t \xrightarrow{d}{}^{*}_{\mathbf{m}} t'\ ])\ ||\ R \text{ is a redex } d \leq D \text{ in } t\ ]$$

Proving it decreases

$t \to_{\mathbf{m}} s$ analyzing each redex contribute

Proving it decreases

$t \to_{\mathbf{m}} s$ analyzing each redex contribute

Measure for t

$(3, [\mathcal{T}_2^{\mathfrak{m}}(t')$ for 3-devs $])$
$(2, [\mathcal{T}_1^{\mathfrak{m}}(t')$ for 2-devs $])$
$(1, [\mathcal{T}_0^{\mathfrak{m}}(t')$ for 1-devs $])$

Proving it decreases

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$(1, [\mathcal{T}_0^{\mathrm{m}}(t') \text{ for 1-devs }])$

Measure for s

$(3, [\mathcal{T}_2^{\mathrm{m}}(s') \text{ for 3-devs }]), (3, [\mathcal{T}_2^{\mathrm{m}}(s') \text{ for 3-devs }])$
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$(1, [\mathcal{T}_0^{\mathrm{m}}(s') \text{ for 1-devs }]), (1, [\mathcal{T}_0^{\mathrm{m}}(s') \text{ for 1-devs }])$

+ pairs, - devs

+ pairs, - devs

+ pairs, + devs

Proving it decreases

$t \to_{\mathbf{m}} s$ analyzing each redex contribute

Measure for t

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+ pairs, - devs

+ pairs, - devs

+ pairs, + devs

Non-erasure + redex replication

Proving it decreases

$t \to_{\mathtt{m}} s$ analyzing each redex contribute

Measure for t

$(3, [\mathcal{T}^{\mathtt{m}}_2(t') \text{ for 3-devs }])$
$(2, [\mathcal{T}^{\mathtt{m}}_1(t') \text{ for 2-devs }])$
$(1, [\mathcal{T}^{\mathtt{m}}_0(t') \text{ for 1-devs }])$

Measure for s

$(3, [\mathcal{T}^{\mathtt{m}}_2(s') \text{ for 3-devs }]), (3, [\mathcal{T}^{\mathtt{m}}_2(s') \text{ for 3-devs }])$
$(2, [\mathcal{T}^{\mathtt{m}}_1(s') \text{ for 2-devs }]), (2, [\mathcal{T}^{\mathtt{m}}_1(s') \text{ for 2-devs }])$
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+ pairs, - devs

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Non-erasure + redex replication

Weight analysis

$$t \xrightarrow[\mathtt{m}]{d} t'$$

$$[\mathcal{T}^{\mathtt{m}}_{d-1}(s) \text{ for } \underbrace{d\text{-devs}}_{X}] \qquad [\mathcal{T}^{\mathtt{m}}_{d-1}(s') \text{ for } \underbrace{d\text{-devs}}_{Y}]$$

Proving it decreases

$t \to_{\mathtt{m}} s$ analyzing each redex contribute

Measure for t

$(3, [\mathcal{T}^{\mathtt{m}}_2(t') \text{ for 3-devs }])$
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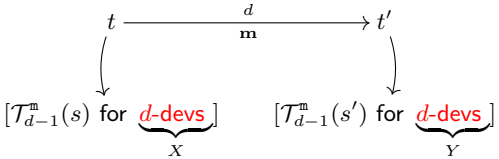
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Non-erasure + redex replication

Weight analysis



Relate developments

Proving it decreases

$t \to_{\mathbf{m}} s$ analyzing each redex contribute

Measure for t

$(3, [\mathcal{T}^{\mathbf{m}}_2(t')$ for 3-devs $])$
$(2, [\mathcal{T}^{\mathbf{m}}_1(t')$ for 2-devs $])$
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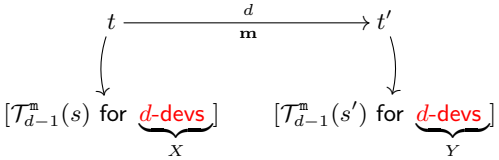
Measure for s

$(3, [\mathcal{T}^{\mathbf{m}}_2(s')$ for 3-devs $]), (3, [\mathcal{T}^{\mathbf{m}}_2(s')$ for 3-devs $])$
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Non-erasure + redex replication

Weight analysis



Relate developments

easy injection $Y \mapsto X$
empty dev $t \to^0_{\mathbf{m}} t$
same targets + 1

$d = D$

Proving it decreases

$t \to_{\mathbf{m}} s$ analyzing each redex contribute

Measure for t

$(3, [\mathcal{T}^{\mathfrak{m}}_2(t')$ for 3-devs $])$
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Measure for s

$(3, [\mathcal{T}^{\mathfrak{m}}_2(s')$ for 3-devs $]), (3, [\mathcal{T}^{\mathfrak{m}}_2(s')$ for 3-devs $])$
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Non-erasure + redex replication

Weight analysis

$$t \xrightarrow[\mathbf{m}]{d} t'$$

$[\mathcal{T}^{\mathfrak{m}}_{d-1}(s)$ for $\underbrace{d\text{-devs}}_{X}]$ $\quad$ $[\mathcal{T}^{\mathfrak{m}}_{d-1}(s')$ for $\underbrace{d\text{-devs}}_{Y}]$

Relate developments

$d = D$

easy injection $Y \mapsto X$
empty dev $t \to^0_{\mathbf{m}} t$
same targets + 1

$d < D$

lifting property
for target in Y
greater target in X

Quick conclusions

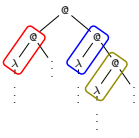
Main theorem

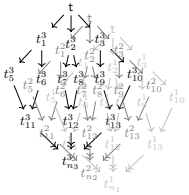
$$M \to_\beta N \qquad \Longrightarrow \qquad \mathcal{T}^{\mathbb{m}}(M) > \mathcal{T}^{\mathbb{m}}(N)$$

How redex degrees could be *complemented* to make the DM strong

Degree-nested multisets

built from degree-restricted reachable terms





$$\left[\ \mathcal{T}_2(t)\ ,\ \mathcal{T}_2(t^3_1)\ ,\ \left[\ \cdots \mathcal{T}_1(t)\ \ \mathcal{T}_2(t^3_{n_3})^2_1\right]\ ,\ \left[\ \mathcal{T}_0(t)\ ,\ \mathcal{T}_0(t^2_1)\ ,\ \mathcal{T}_0(t^2_{n_2})\ \right]\ ,\ \cdots\ ,\ \mathcal{T}_0(t^1_{n_1})\ \right]$$

$$\mathcal{T}_3(t) = [\ (3,\ ),\ (2,\ ),\ (1,\ )\ ]$$
$$\mathcal{T}_2(t) = [\ (2,\ )\ ,\ (1,\ )\ ]$$
$$\mathcal{T}_1(t) = [\ (1,\ )\ ]$$
$$\mathcal{T}_0(t) = [\ \ ]$$

Thoughts on complexity

- *Easeness* dimensions
- Reducibility "opposite"
- Future distillation

Conclusions and future work

Contributions of the thesis to the field

Contributions of the thesis to the field

Advance of syntactic proofs of SN in typed λ–calculus

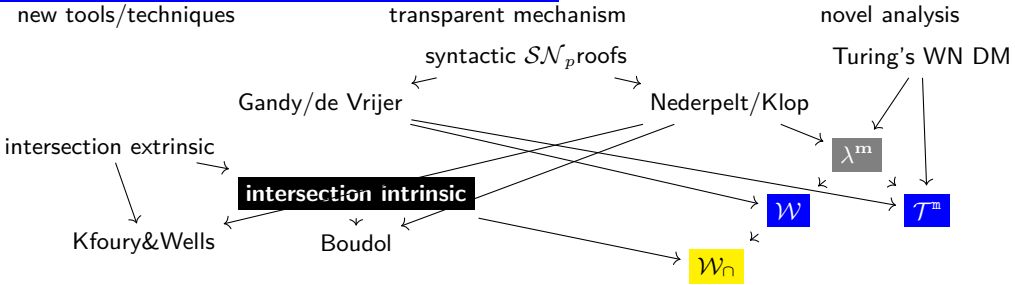
Contributions of the thesis to the field

Advance of syntactic proofs of SN in typed λ–calculus

new tools/techniques | transparent mechanism | novel analysis

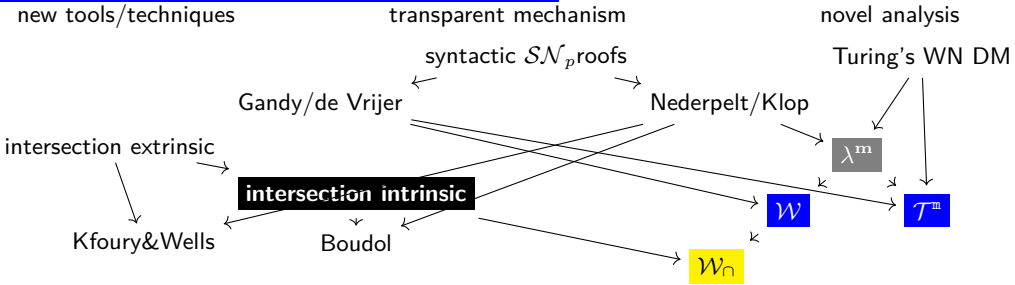
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Contributions of the thesis to the field

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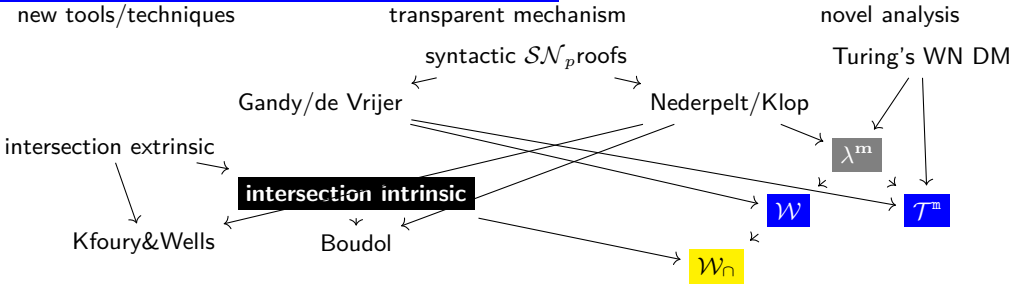


The koan: the ultimate, ambitious goal

- remains unsolved
- remains as motivation
- valuable advances towards

Contributions of the thesis to the field

Advance of syntactic proofs of SN in typed λ–calculus



The koan: the ultimate, ambitious goal

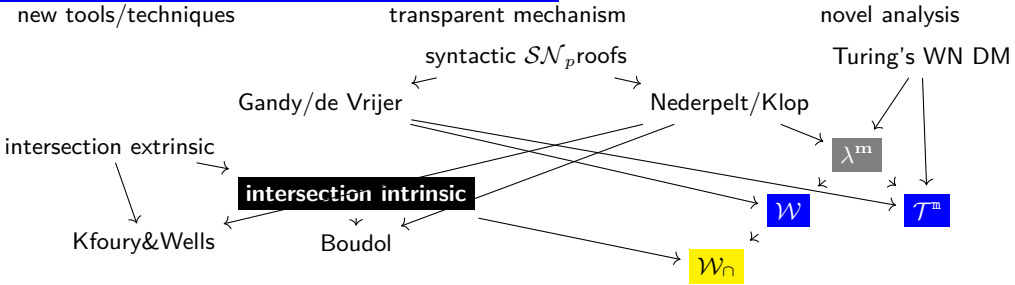
- remains unsolved
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- valuable advances towards

Continuing this line of work

- capture SN trails into syntactic props
- insight on big question behind koan and my thesis

Contributions of the thesis to the field

Advance of syntactic proofs of SN in typed λ–calculus



The koan: the ultimate, ambitious goal

- remains unsolved
- remains as motivation
- valuable advances towards

Continuing this line of work

- capture SN trails into syntactic props
- insight on big question behind koan and my thesis

What lies behind execution that makes it become closer to termination at each step?

Future work

Deepening

Look for insight in defined measures

New measures	exact $\mathcal{W}_\cap$	simplified $\mathcal{T}^{\mathbb{m}}$
Further (formal) analysis	$\mathcal{W}$ and Gandy/deVrijer's	$\mathcal{W}$ and CT
Mechanize in proof assistants	$\mathcal{W}$, $\mathcal{W}_\cap$, $\hat{\mathcal{W}}$, $\hat{\mathcal{W}}_\cap$	$\mathcal{T}^{\mathbb{m}}$

Expanding

Test the limits of $\mathcal{W}$

Other IIT	HN characterization	untyped subterms allowed
System F	sounds impossible due to impred.	there are antecedents
Modulo isomorphisms	neutrality instability	RED proofs very hard

Thanks!

Observations

The adaptation $\mathcal{W}_\cap$ is conservative?

Yes, noting that...

new cases + modified existing cases

Observations

Γ in $\Lambda_{\cap}^{\mathsf{Ch}}$

For true correspondence

decorating + erasing giving same derivation

Observations

snoc operation in $\mathtt{L}$ and equivalence relation *e.g.*, $t \rightleftarrows t\Box$

$\mathtt{L}$ is meta-notation — easily handle redex cases — have all relevant information at once

$$\mathtt{S}_D((\lambda x^A.t)s) := \mathtt{S}_D(t)[\mathtt{S}_D(s)/x^A]\langle \mathtt{S}_D(s)\rangle \quad \text{(with deg } = D)$$

Observations

On using the exact measure to assess complexity

Theoretical
Accuracy and elegance

Utility
Today expensive, future?

Observations

How is $\mathcal{W}$ direct for SN? Does not it reduce SN to WN too?

CT+WDM: DM $\implies$ WN

SN from CT

Insight: we managed to put SN information inside the delivered value (vs WN info)

Ours implies SN directly

Koan: find something decreasing for all reductions, not some

Observations

Orthogonality of HRS does not imply restricted reduction stay orthogonal

True | Changed after peer-reviews | Original proof through parallel reductions

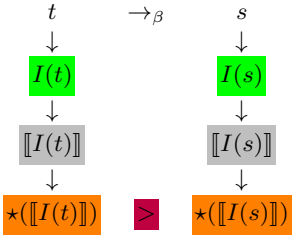
What others looked? Existing decreasing measures

Gandy

- **translation** to λ-I
- **interpretation** of terms into HSMF: hereditarily strictly monotonic functions
- **projection** of HSMF into natural numbers
- **proof** based on HSMF properties

Derivatives

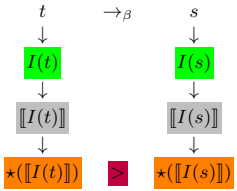
- de Vrijer: simplification
- de Vrijer: refinement to exactness



Related works I

Gandy

- **translation** to λ-I
- **interpretation** into hereditarily strictly monotonic functions
- **projection** of HSMF into $\mathbb{N}$
- **proof** based on HSMF properties

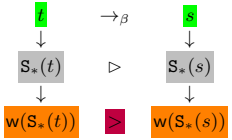


Comparison

- General analysis and construction is different
- But are they deeply related?
- A precise correspondence has not been established and is not immediate

Ours

- **extension** to $\lambda^{\mathbf{m}}$
- **normal form** calculation in $\lambda^{\mathbf{m}}$
- **projection** into $\mathbb{N}$
- **proof** based on λ-calculi properties



Some clear differences

- one less step
- domain: HSMF (math) vs λ
- non-erasure flavour λ-Ivs $\lambda^{\mathbf{m}}$
- obtained value
- proof: HSMF props. (mono.) vs λ props. (CR, INC, WN, comm., tof)

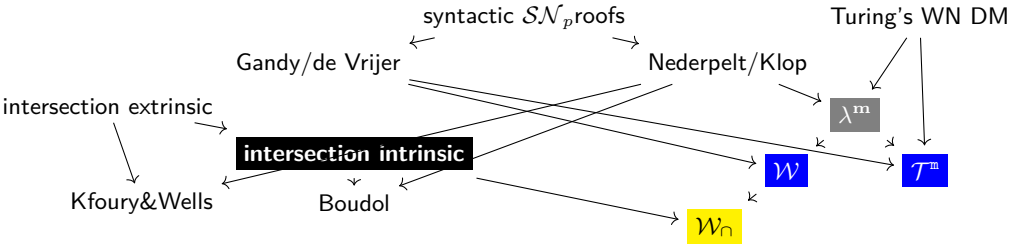
Related works II

The Conservation Theorem

- Non-erasure to reduce SN to WN
- SN through weak DM (implying WN) + CT
- WDM usually following Turing's

Our approach

- CT holds from non-erasure, but is not used
- $\mathcal{W}$ is a strong DM, it alone implies SN



Related works I

Reduction of SN to WN + Redex-degree weak DM

[Kfoury & Wells'95]

- **Codomain:** **multiset of $\mathbb{N}$**
- **Type:** **weak** (indirect for SN)
- **Calculus:** à la Church, ad hoc

[Boudol'03]

- **Codomain:** **pair of $\mathbb{N}$**
- **Type:** **weak** (indirect for SN)
- **Calculus:** à la Curry

Related works I

Reduction of SN to WN + Redex-degree weak DM

[Kfoury & Wells'95]

- **Codomain:** **multiset of** $\mathbb{N}$
- **Type:** **weak** (indirect for SN)
- **Calculus:** à la Church, ad hoc

[Boudol'03]

- **Codomain:** **pair of** $\mathbb{N}$
- **Type:** **weak** (indirect for SN)
- **Calculus:** à la Curry

Our proposal

- **Codomain:** $\mathbb{N}$
- **Type:** **strong** (direct for SN)
- **Calculus:** à la Church, **proved**

Related works II: A translation based measure

Translation from IIT to STLC: $\mathcal{A}_\cap$ [Bucciarelli, Piperno & Salvo]

$$\frac{}{x : \{A \to A, A\} \vdash x\,x : A} \quad \rightsquigarrow \quad \frac{}{y : A \to A, z : A \vdash y\,z : A}$$

$$\mathcal{A}_\cap((\lambda x.xx)(\lambda y.y)) = (\lambda x_1.\lambda x_2.x_1x_2)(\lambda y.y)(\lambda y.y)$$

Measure by composition: $\mathcal{A}_\cap \circ \#$

Strength Straigthforward: composing proved results

Flaw Misses the intersection type details

Issues

- Moves problem to different domain
- Same reduction, different terms
- Intersection type information is lost

Example

$$(\lambda x.xx)(\lambda y.y) \to_\beta (\lambda y.y)(\lambda y.y)$$

- No analysis on computing process of IIT exclusive terms
- Hard limitation for further analysis
  e.g., impossible to refine to exactness

$$\begin{aligned}\mathcal{A}_\cap((\lambda x.xx)(\lambda y.y)) &= (\lambda x_1.\lambda x_2.x_1x_2)(\lambda y.y)(\lambda y.y)\\ &\to_\beta (\lambda x_2.(\lambda y.y)x_2)(\lambda y.y)\\ &\to_\beta (\lambda y.y)(\lambda y.y)\end{aligned}$$

Exactness: the proof

Arithmetic decomposition of $\hat{\mathcal{W}}$

$$\hat{\mathcal{W}}(t\langle s\rangle u) = \hat{\mathcal{W}}((tu)\langle s\rangle)$$

Proof

- when $\mathtt{S}_*(t\langle s\rangle u)$ becomes a d-redex through simplification?
- t becomes λ under $\mathtt{S}_D(t)$ with $D > d$
- $\mathtt{S}_d$ preserves structure from D to $d+1$
- once t became λ, both sides give the same result

$$\mathtt{S}_{\leq d}((\lambda x.t')\langle \mathtt{S}_{>d}(s)\rangle \mathtt{S}_{>d}(u)) \qquad \mathtt{S}_{\leq d}((\lambda x.t')\mathtt{S}_{>d}(u)\langle \mathtt{S}_{>d}(s)\rangle)$$

$$\mathtt{S}_{<d}(\mathtt{S}_d(t')\langle \mathtt{S}_{\geq d}(u)\rangle\langle \mathtt{S}_{\geq d}(s)\rangle)$$

WIP: refining $\mathcal{W}_\cap$ (IIT) to exactness

Extra motivation There is no decreasing measure giving an exact upper bound

Extra problems Curry vs Church: the reduction gap induced by the translation

$$\dfrac{\dfrac{}{x : \{A \to A, A\} \vdash x : A}}{\vdash \lambda x.x : \{A \to A, A\} \to A} \quad \dfrac{\vdots \quad \vdots}{\Vdash II : \{A \to A, A\}}$$

$$\vdash (\lambda x.x \underset{\text{dec.}}{)}(II)$$

$$(\lambda x.x)(II) \xrightarrow{\beta} II \xrightarrow{\beta} I$$

$$(\lambda x^{\{A\to A, A\}}.x^A)\{(II)^{A\to A}, (II)^A\} \xrightarrow{\mathsf{Ch}} (\lambda x^{\{A\to A, A\}}.x^A)\{I^{A\to A}, (II)^A\} \xrightarrow{\mathsf{Ch}} (\lambda x^{\{A\to A, A\}}.x^A)\{I^{A\to A}, I^A\} \xrightarrow{\mathsf{Ch}} I^A$$

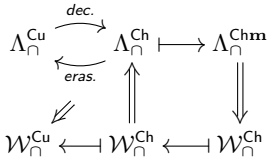
$$(\lambda x^{\{A\to A, A\}}.x^A)\{(II)^{A\to A}, (II)^A\}$$

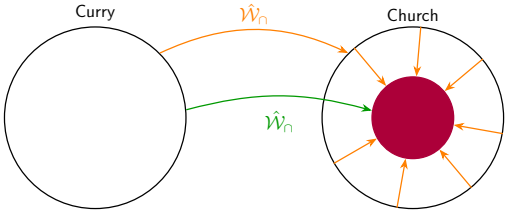
- One to one impossible
  $\{t_1, ..., t_i, ...t_n\} \to_{\mathsf{Ch}}$
  $\{t_1, ..., t'_i, ..., t_n\}$
- Enough $F_\infty(M) = F^\cap_\infty(t^M)$
- When this breaks?

Extra typing information $\implies$ **Extra argument replication** $\implies$ **Extra reduction steps**

Two paths

Objective Avoid extra information in derivations





New system restricting info

- forbid derivations with extra information
- recreate the work from $\mathcal{W}_\cap$ there
- *e.g.*, multiplicative presentation of IIT [De Benedetti & Ronchi Della Rocca]

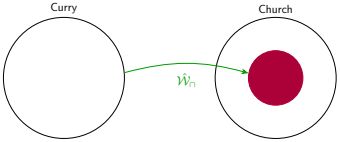
$$F_\infty(M) = F^\cap_\infty(t^M)$$

Same $\Lambda^{\mathsf{Ch}}_\cap$ filtering redundancy

- discard *bad* derivations
- reuse the results
- shrink arbitrary derivations and compute measure on pure ones

$$\text{pure } \Pi = \Gamma \vdash M : A \implies F_\infty(M) = F^\cap_\infty(\mathit{dec}(\Pi))$$

Path 2: filter pure derivations



$$\Lambda_\cap^{\mathsf{Cu}} \xrightarrow{\textit{pure}} \Lambda_\cap^{\mathsf{Cu}} \underset{\textit{eras.}}{\overset{\textit{dec}}{\rightleftarrows}} \Lambda_\cap^{\mathsf{Ch}}$$

$$\hat{\mathcal{W}}_\cap(M) = \mathcal{W}_\cap(\textit{dec}(\textit{pure}(M)))$$

Weakening is twofold in IIT

$$\frac{}{x : \{A\}, y : \{B\} \vdash x : A}$$

$$\frac{\dfrac{}{x : \{A \to A, A\} \vdash x : A}}{\vdash \lambda x.x : \{A \to A, A\} \to A}$$

Canonical derivation

1. variables in Γ occur in term
2. non-erased variables have no unused type
3. erased variables have one type

Some properties

- each derivation has a corresponding canonical
- distinct derivations of a single term share the canonical: if Π_1 and Π_2 derive A for M then they share canonical

Assuming *pure* defined:

1. Prove soundness
   pure(Π) valid and canonical
2. Prove canonicity implies
   $F_\infty(M) = F_\infty^\cap(\textit{dec}(\Pi))$

pure: the canonicalization function

Naive attempt

$$\frac{}{x : \{A_1, ..., A_n\}, y : \{B\} \vdash y : B}$$
$$\overline{x : \{A_1, \cancel{..., A_n}\} \vdash \lambda x.y : \{A\} \to B}$$

$$\frac{}{x : \{A \to A, A\} \vdash x : A}$$
$$\overline{\vdash \lambda x.x : \{\cancel{A \to A}, A\} \to A}$$

Weakenings to address

- Unused types in bound vars
- Function types in variables

Weakening is spread

$$\frac{\dfrac{\dfrac{}{x : \{A, B\} \to A} \quad \dfrac{}{N : \{A, B\}}}{xN : A}}{\lambda x.xN : \{A, B\} \to A} \qquad \frac{}{y : \{A, B\} \vdash y : A}$$
$$\overline{(\lambda x.xN)(\lambda y.y) : A}$$

Latent substitute argument eventually replacing variable

Specialization

- consistently changing the type of a bound variable
- based on the canonical of its latent substitute
- spreading to variable arguments

Idea *pure* by recursion

1. Gather specializations from arguments $\{A, B\} \to A$ becomes $\{A\} \to A$ for $\lambda y.y$
2. Push argument's spec to stack at @
3. Pop spec at corresponding λ and spread $x : \{A, B\} \to A$ becomes $x : \{A\} \to A$ $N : \{A, B\}$ becomes $N : \{A\}$

Measure $\mathcal{T}$: first attempt

Example

$$\underbrace{(\lambda x^{\tau}....x...x...)\underbrace{(K\underbrace{(\underbrace{Iy^{\tau}}_{1})}_{2}(y^{\tau}))}_{1}}_{1} \quad \rightarrow_{\beta} \quad ...(K\underbrace{(\underbrace{Iy^{\tau}}_{1})}_{2}(y^{\tau}))...\underbrace{(K\underbrace{(\underbrace{Iy^{\tau}}_{1})}_{2}(y^{\tau}))}_{2}...$$

$$\mathcal{T}_2(t) \;=\; [\ (2,\ )\ ,\ (1,[])\ ,\ (1,[])\ ]$$

$$\mathcal{T}_1(t) \;=\; [\ (1,[])\ ,\ (1,[])\ ]$$

$$\mathcal{T}_0(t) \;=\; [\ ]$$

$$\mathcal{T}_2(t') \;=\; [\ (2,\ )\ ,\ (2,\ )\ ,\ (1,[])\ ,\ (1,[])\ ]$$

$$\mathcal{T}_1(t') \;=\; [\ (1,[])\ ,\ (1,[])\ ]$$

$$\mathcal{T}_0(t') \;=\; [\ ]$$

Developments by degree

Development of a set of redexes

Reduction seq. built from redexes or residuals of set $\{R, S, T\} \in t$ $\quad t \to_R t_1 \to_{S'} t_2 \to_{S'} t_3$

Reduction by degrees

$t \xrightarrow{d} s$ contracted redex has degree d

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Reduction by degrees

$t \xrightarrow{d} s$ contracted redex has degree d

Developments by degrees: d-devs

Devs. of sets of d-redexes $\{R^2, S^2, \not{T}^3\} \in t$ $\quad t \xrightarrow{2}_R t_1 \xrightarrow{2}_{S'} t_2 \xrightarrow{2}_{S'} t_3$

Developments by degree

Development of a set of redexes

Reduction seq. built from redexes or residuals of set

$$\{R, S, T\} \in t \qquad t \to_R t_1 \to_{S'} t_2 \to_{S'} t_3$$

Reduction by degrees

$t \xrightarrow{d} s$ contracted redex has degree d

Developments by degrees: d-devs

Devs. of sets of d-redexes

$$\{R^2, S^2, \not{T^3}\} \in t \qquad t \xrightarrow{2}_R t_1 \xrightarrow{2}_{S'} t_2 \xrightarrow{2}_{S'} t_3$$

All d-devs

all paths of the complete d-reduction graph from t

Developments by degree

Development of a set of redexes

Reduction seq. built from redexes or residuals of set

$$\{R, S, T\} \in t \qquad t \to_R t_1 \to_{S'} t_2 \to_{S'} t_3$$

Reduction by degrees

$t \xrightarrow{d} s$ contracted redex has degree d

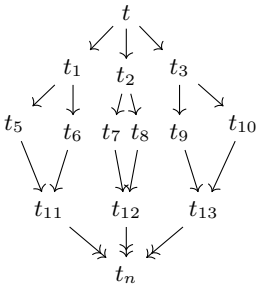
Developments by degrees: d-devs

Devs. of sets of d-redexes

$$\{R^2, S^2, \cancel{T^3}\} \in t \qquad t \xrightarrow{2}_R t_1 \xrightarrow{2}_{S'} t_2 \xrightarrow{2}_{S'} t_3$$

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Developments by degree

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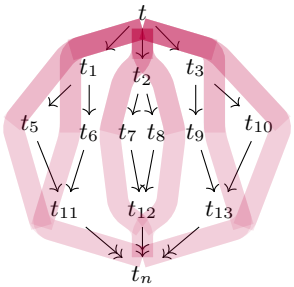
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Developments by degree

Development of a set of redexes

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Reduction by degrees

$t \xrightarrow{d} s$ contracted redex has degree d

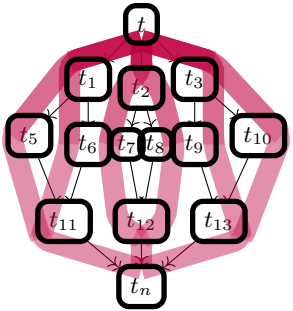
Developments by degrees: d-devs

Devs. of sets of d-redexes

$$\{R^2, S^2, \cancel{T^3}\} \in t \qquad t \xrightarrow{2}_R t_1 \xrightarrow{2}_{S'} t_2 \xrightarrow{2}_{S'} t_3$$

All d-devs

all paths of the complete d-reduction graph from t



Proving it decreases

Check what happens to all the other redexes after a contraction

Low/decrease lemma

$$t \xrightarrow{D}_{\mathbf{m}} t' \implies \mathcal{T}^{\mathbb{m}}_d(t) \prec \mathcal{T}^{\mathbb{m}}_d(t')$$

$$d \leq D \wedge t \xrightarrow{D}_{\mathbf{m}} t' \implies \mathcal{T}^{\mathbb{m}}_{\leq D}(t) \preceq \mathcal{T}^{\mathbb{m}}_{\leq D}(t')$$

Proof

$$\begin{aligned} & \mathcal{T}^{\mathbb{m}}_{1..D}(t) \\ = \; & \mathcal{T}^{\mathbb{m}}_{1..(d-1)}(t) + \mathcal{T}^{\mathbb{m}}_{d}(t) + \mathcal{T}^{\mathbb{m}}_{(d+1)..D}(t) \\ \succeq \; & \cancel{\mathcal{T}^{\mathbb{m}}_{1..(d-1)}(t)} + \mathcal{T}^{\mathbb{m}}_{d}(t) + \mathcal{T}^{\mathbb{m}}_{(d+1)..D}(t) \end{aligned}$$

How each redex in the term contribute something to the measure

$\mathcal{T}^{\mathbb{m}}_d(t)$: $[(d, \mathcal{R}_d(t)) \mid\mid R\ d\text{-redex in } t]$

- $(d, \mathcal{R}_d(t))$ contributed is now missing
- what about other d-redexes *weight*?

$\mathcal{T}^{\mathbb{m}}_{(d+1)..D}(t)$: $[(j, \mathcal{R}_d(t)) \mid\mid R\ j\text{-redex in } t, j \in (d+1)..D]$

- might have more j-redexes due to copy
- but all with a lighter *weight*

Weight analysis

$$d \leq D \wedge t \xrightarrow{D}_{\mathbf{m}} t' \implies \mathcal{R}_D(t) \prec \mathcal{R}_D(t')$$

$$\mathcal{R}_D(t) \prec \mathcal{R}_d(t')$$

Weight analysis

$$d \leq D \wedge t \xrightarrow{D}_{\mathbf{m}} t' \implies \mathcal{R}_D(t) \prec \mathcal{R}_D(t')$$ **Relate developments**

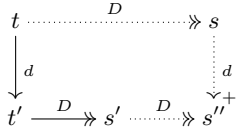
Same degree $D = d$ $\quad R : t \xrightarrow{d}_{\mathbf{m}} t' \qquad X = \{\sigma \mid \sigma : t \xrightarrow{d}{}^{*}_{\mathbf{m}} s\} \qquad Y = \{\rho \mid \rho : t' \xrightarrow{d}{}^{*}_{\mathbf{m}} s'\}$

1. Define injective function from Y to X by preppending R: $\quad \varphi(\rho) = R \circ \rho$
2. Targets are the same $\sigma^{\mathsf{tgt}} = \varphi(\rho)^{\mathsf{tgt}}$ $\quad$ all targets measured in $\mathcal{R}_d(t')$ are measured in $\mathcal{R}_d(t)$
3. Empty dev $t \xrightarrow{d}{}^{0}_{\mathbf{m}} t \in X$ $\quad$ there is target measured in $\mathcal{R}_d(t)$ and not in $\mathcal{R}_d(t')$

$$\mathcal{R}_d(t) \prec \mathcal{R}_d(t')$$

Greater degree $D > d$ $\quad R : t \xrightarrow{d}_{\mathbf{m}} t' \qquad X = \{\sigma \mid \sigma : t \xrightarrow{D}{}^{*}_{\mathbf{m}} s\} \qquad Y = \{\rho \mid \rho : t' \xrightarrow{D}{}^{*}_{\mathbf{m}} s'\}$

Non-erasure enables lifting



1. We can take the d-step after D-steps
2. Assign ρ to the σ enabling .. $\xrightarrow{d}{}^{+}_{\mathbf{m}} s''$
3. $\mathcal{R}$ is built from target's lower measures
4. $s \xrightarrow{d}_{\mathbf{m}} s''$ so by IH on d: $\mathcal{T}^{\mathbf{m}}_{\leq D-1}(s) \succ \mathcal{T}^{\mathbf{m}}_{\leq D-1}(s)''$
5. $s' \xrightarrow{D}_{\mathbf{m}} s''$ so by non-erasure: $\mathcal{T}^{\mathbf{m}}_{\leq D-1}(s') \preceq \mathcal{T}^{\mathbf{m}}_{\leq D-1}(s'')$

For each dev in X, we found a dev in Y with a greater measured target, hence $\mathcal{R}_D(t) \prec \mathcal{R}_D(t')$

RED Gallier/dV